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**MatchScore
Software Requirements Specification
Version 01.00**

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Software Requirements Specification

1. Introduction

The introduction of this Software Requirements Specification (SRS) provides an overview of the entire document, including its purpose, scope, definitions, references, and structure. This document defines the requirements for **MatchScore**, a web platform that allows users to build resumes and portfolios, score them against job postings using a deterministic text-similarity algorithm, and receive AI-generated explanations and edit recommendations to improve their fit for a specific role.

1.1 Purpose

The purpose of this document is to define the functional and non-functional requirements of MatchScore for stakeholders, developers, and evaluators. It describes what the system does, how users interact with it, and the constraints under which it operates, serving as a reference throughout the software development lifecycle.

1.2 Scope

MatchScore allows users to create and edit resumes and portfolio content using templates, export/import documents as PDF, and compare their resume against a pasted job posting to receive a computed match score. The system uses a deterministic algorithm (TF-IDF and cosine similarity) to generate the score, and an AI layer to explain the score and suggest specific edits. The scope of this project covers the resume builder, scoring engine, and AI guidance features; it does not cover job-posting scraping/auto-import or a full multi-page portfolio site builder in its initial version.

1.3 Definitions, Acronyms, and Abbreviations

Term	Description
SRS	Software Requirements Specification
TF-IDF	Term Frequency-Inverse Document Frequency, a text-weighting technique
Cosine Similarity	A measure of similarity between two text vectors
WYSIWYG	“What You See Is What You Get” – An editor that shows final formatting while editing
ATS	Applicant Tracking System, software used by employers to filter resumes
LLM	Large Language Model
MVP	Minimum Viable Product

1.4 References

- TF-IDF and Cosine Similarity documentation (scikit-learn)
- Free-tier LLM API documentation (Google Gemini / Groq)
- SRS Template provided for this course

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1.5 Overview

The remaining sections of this document describe the overall system, its users, and its specific functional and non-functional requirements. Section 2 gives a general description of MatchScore. Section 3 details the specific functional requirements (organized by feature), usability, reliability, performance, security, design constraints, and interfaces.

2. Overall Description

MatchScore addresses a common problem for job seekers: resume-building tools and resume-scoring tools are typically separate products, requiring users to manually copy feedback from one tool into another. MatchScore combines both into a single platform — users build their resume/portfolio, paste a job posting, receive a deterministic match score with an AI-generated explanation, and can apply AI-suggested edits directly within the same editor. The system's primary stakeholders are individual job seekers, particularly students and early-career professionals.

The system's core functions include: template-based document creation, live in-browser editing, PDF import/export, job-fit scoring, and AI-driven explanation and recommendation generation. The score itself is always computed deterministically (never AI-generated) to ensure the result is reproducible and explainable — the AI's role is strictly limited to interpreting and acting on that score.

3. Specific Requirements

3.1 Functionality

3.1.1 *User Registration & Authentication*

3.1.1.1 The system shall allow users to create an account using email and password.

3.1.1.2 The system shall allow users to log in and log out of their account.

3.1.1.3 The system shall associate all documents, templates, and match results with the logged-in user's account.

3.1.2 *Template System*

3.1.2.1 The system shall provide a library of pre-built resume and portfolio templates.

3.1.2.2 The system shall allow users to manually create a custom template by arranging sections, fonts, and colors.

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The system shall allow users to generate a template using AI by describing a desired style or purpose.

3.1.2.3 The system shall restrict AI-generated templates to layout and styling only, not document content.

3.1.3 **Live Web Editor**

3.1.3.1 The system shall provide a WYSIWYG editor for editing resume and portfolio content directly in the browser.

3.1.3.2 The system shall autosave user edits without requiring manual save actions.

3.1.3.3 The system shall organize editable content into distinct sections (e.g., Experience, Skills, Education).

3.1.4 **PDF Export**

3.1.4.1 The system shall allow users to export their completed resume as a PDF file matching the selected template.

3.1.5 **PDF Import**

3.1.5.1.1 The system shall allow users to upload an existing PDF resume.

3.1.5.1.2 The system shall extract text from the uploaded PDF and populate it into an editable template.

3.1.5.1.3 The system shall display extracted content for user review before saving.

3.1.6 **Job Posting Input**

3.1.6.1 The system shall allow users to paste job posting text into a designated input field.

3.1.6.2 The system shall reject input below a minimum length threshold to prevent invalid submissions.

3.1.7 **Job-Fit Scoring Engine**

3.1.7.1 The system shall preprocess resume and job posting text (tokenization, stopword removal, stemming).

3.1.7.2 The system shall compute a match score using TF-IDF vectorization and cosine similarity.

3.1.7.3 The system shall identify and display matched and missing key terms between the resume

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and job posting.

3.1.7.4 The system shall produce the same score for the same resume/job posting pair every time (deterministic behavior).

3.1.8 ***AI Score Explanation***

3.1.8.1 The system shall generate a plain-language explanation of the computed match score using AI, based on the score and extracted terms.

3.1.9 ***AI Auto-Edit***

3.1.9.1 The system shall allow users to apply an AI-generated suggestion directly to the relevant section of their document.

3.1.9.2 The system shall restrict AI auto-edits to rephrasing or reorganizing existing user content, and shall not introduce new factual claims without explicit user confirmation.

3.1.9.3 The system shall allow users to undo an applied AI edit.

3.1.10 ***Score History***

3.1.10.1 The system shall store past match results and allow users to view how their score has changed over time.

3.1.11 ***AI Edit Recommendations***

3.1.11.1 The system shall generate specific, section-targeted edit suggestions using AI, based on the missing terms and job posting context.

3.2 **Usability**

3.2.1 ***Graphical User Interface***

- The system shall provide a consistent visual layout across all pages of the application.
- The system shall provide clear visual indicators for AI-generated content versus user-authored content.

3.2.2 ***Accessibility***

- The system shall support standard browser accessibility features (keyboard navigation, screen-reader-compatible labels).

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3.3 Reliability & Availability

3.3.1 *AI Service Fallback*

- The system shall display a graceful error message if the AI API is unavailable or a rate limit is reached, rather than failing silently.

3.3.2 *Data Persistence*

- The system shall save user documents and match results to a persistent database, preventing data loss between sessions.

3.4 Performance

- The system shall return a computed match score within a reasonable time (target: under 5 seconds) after a job posting is submitted.
- The system shall cache AI-generated explanations and recommendations per resume/job-posting pair to reduce redundant API calls.

3.5 Security

3.5.1 *Data Transfer*

- The system shall use HTTPS for all data transmitted between the client and server.
- The system shall not expose API keys or credentials to the client.

3.5.2 *Data Storage*

- The system shall store user passwords using a secure hashing algorithm, never in plaintext.
- The system shall restrict access to a user's documents and match results to that user's authenticated account only.

3.6 Design Constraints

3.6.1 *Free-Tier AI API*

- The system shall be built to operate within the rate limits of a free-tier LLM API (e.g., Google Gemini or Groq).
- The system shall minimize live AI calls by pre-generating or caching content where feasible.

3.6.2 *Web-Based Platform*

- The system shall be accessible through standard modern web browsers without requiring installation.

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3.7 Interfaces

3.7.1 *User Interfaces*

- The user interface shall be compatible with modern browsers (Chrome, Firefox, Edge).
- The user interface shall be built using React.

3.7.2 *Software Interfaces*

- The system shall communicate with a free-tier LLM API for score explanation, recommendation generation, and template generation.
- The system shall use a PDF generation library to export documents.
- The system shall use a PDF parsing library to extract text from imported documents.

3.7.3 *Communications Interfaces*

- The system shall use the HTTPS protocol for all client-server communication.